Xin Dong

Postdoctoral Researcher, Ph.D. in Bioinformatics

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RESEARCH APPOINTMENTS

Jul. 2026 – Present	Postdoctoral Researcher	Department of Chemistry, University of Chicago Advisor: Prof. Chuan He & Prof. Mengjie Chen
Sep. 2025 – Jul. 2026	Research Scientist	Sycamore Research Institute of Life Sciences, Shanghai
May 2025 – Sep. 2025	Bioinformatician (Part-time)	GV20 Therapeutics LLC
Jul. 2024 – Sep. 2025	Research Assistant	School of Life Sciences and Technology, Tongji University

EDUCATION

Sep. 2018 – Jul. 2024 **Ph.D. in Bioinformatics**
School of Life Sciences and Technology, Tongji University, China
Advisors: Prof. Chenfei Wang & Prof. Xiaole Shirley Liu

Sep. 2014 – Jun. 2018 **B.S. in Biotechnology**
College of Life Sciences, Shandong Normal University, China

RESEARCH EXPERIENCE

Research Interests

RNA-mediated transcriptional regulation; RNA-chromatin interactions; AI for genomics; single-cell and spatial multiomics.

Postdoctoral Researcher, Department of Chemistry, University of Chicago (Jul. 2026 – Present)

Advisor: Prof. Chuan He & Prof. Mengjie Chen

- Develop computational frameworks integrating RNA-associated molecular data and epigenomic profiles to investigate RNA-mediated transcriptional regulation.
- Investigate the regulatory functions of small nuclear RNA (snRNA) variants and their associations with gene expression programs.
- Analyze FTO perturbation and RNA modification datasets to characterize RNA-dependent regulatory responses.

Research Scientist, Sycamore Research Institute of Life Sciences (Sep. 2025 – Jul. 2026)

- Conducted computational analyses for studies of small RNAs and glycosylated RNAs (glycoRNAs) (*Under review*).
- Characterized the roles of YTHDF family proteins in myeloid cells in the context of immunotherapy (*Under review*).
- Constructed RNA-RNA interaction maps in collaboration with experimentalists.

Bioinformatician, GV20 Therapeutics LLC (May 2025 – Sep. 2025)

- Obtained a Good Clinical Practice (GCP) training certificate and contributed to establishing a trial master file (TMF) management system.

Research Assistant, School of Life Sciences and Technology, Tongji University (Jul. 2024 – Sep. 2025)

- Developed PyTorch-based deep learning frameworks for predicting epigenetic states and regulatory information using single-cell/spatial epigenomics data and sequence information.
- Investigated regulatory mechanisms during murine cardiac development through analysis of single-cell multi-omics data (*Under Revision*).

Graduate Student, Tongji University (Sep. 2018 – Jul. 2024)

- **Developed and applied computational methods to analyze single-cell and spatial data**

Advisors: Prof. Chenfei Wang & Prof. Xiaole Shirley Liu

- Developed SCRIP (**S**ingle-**C**ell **R**egulatory Network **I**nference using Ch**IP**-seq and motif) to integrate single-cell chromatin accessibility profiles with large-scale transcriptional regulator ChIP-seq data, enabling inference of regulator activity and target genes.
- Co-developed SCRIPro to extend regulatory network inference to single-cell and spatial multiomic data. The methods outperformed motif-based approaches in assessing TR binding activity and demonstrated their utility in lineage tracing, cell-type clustering, and inference of cell-type-specific GRNs across diverse biological systems.
- Co-developed TISCH (**T**umor **I**mmune **S**ingle **C**ell **H**ub) and TISCH2, integrating approximately 6 million cells from 190 tumor datasets across 50 cancer types to support comparative analyses of the tumor microenvironment. Contributed to data collection and processing, implementation of analysis modules, and website development.

- **Large-scale genomic data mining and integration analysis**
  Advisors: Prof. Xiaole Shirley Liu & Prof. Clifford Meyer
  - Co-developed a GEO data parser and the Snakemake-based CHIPS (**CH**romatin enr**I**chment **P**roce**S**sor) () pipeline to automate data retrieval, metadata annotation, and processing of epigenomic sequencing data
  - Co-developed and maintained the Cistrome Data Browser, providing access to public epigenomic datasets, quality-control metrics, and downstream analysis results.

PUBLICATIONS

2,332 citations | h-index: 11 *Google Scholar, October 27, 2026*

** denotes the Co-first Authors and Hashtag; # denotes the Corresponding Authors.*

(Co)First-Author Papers:

- **Dong, X.***, Tang, K.*, Xu, Y., Wei, H., Han, T. & Wang, C. # Single-cell gene regulation network inference by large-scale data integration. *Nucleic Acids Research*, 50, e126 (2022).
- Sun, F.*, Li, H.*, Sun, D.*, Fu, S.*, Gu, L.*, Shao, X.*, Wang, Q.*, **Dong, X.***, Duan, B.*, Xing, F.*, Wu, J.*, Xiao, M.#, Zhao, F.#, Han, J.-D. J.#, Liu, Q.#, Fan, X.#, Li, C.#, Wang, C.# & Shi, T.# Single-cell omics: experimental workflow, data analyses and applications. [Review] *SCIENCE CHINA Life Sciences* 68, 5–102 (2025).
- Chang, Z.*, Xu, Y.*, **Dong, X.***, Gao, Y. & Wang, C.# Single-cell and spatial multiomic inference of gene regulatory networks using SCRIPro. *Bioinformatics* 40, btae466 (2024).
- Han, Y.*, Wang, Y.*, **Dong, X.***, Sun, D., Liu, Z., Yue, J., Wang, H., Li, T.# & Wang, C.# TISCH2: expanded datasets and new tools for single-cell transcriptome analyses of the tumor microenvironment. *Nucleic Acids Research*, 51, D1425-D1431 (2023).
- Zheng, R.*, **Dong, X.***, Wan, C., Shi, X., Zhang, X.# & Meyer, C.A.# Cistrome Data Browser and Toolkit: analyzing human and mouse genomic data using compendia of ChIP-seq and chromatin accessibility data. *Quantitative Biology*, 8, 267-276 (2020).

Other Papers:

- Tang, K.*, Han, Y.*, Sun, D., **Dong, X.**, Han, T., Wei, H., Shao, W., Hu, J., Liu, Z., Zhang, L., Li, T., Zhang, P., Wu, Q.# & Wang, C#. Reference-guided computational framework identifies microenvironment metabolic subtypes and targets using pan-cancer single-cell datasets. *Genome Med*icine 17, 150 (2025).
- Jiang, R., Chen, J., Pei, L., Huang, S., Feng, M., Ouyang, Z., Yuan, M., Zhu, Y., Yao, S., He, F., Zhang, H., **Dong, X.**, Xie, P.# & Wei, K.# Myosin inhibition enhances cardiomyocyte cell cycle activity through SIRT1-NFAT-mediated H3K9me3 modification. *European Journal of Cell Biology* 104, 151504 (2025).
- Ren, P.*, Shi, X.*, Yu, Z., **Dong, X.**, Ding, X., Wang, J., Sun, L., Yan, Y., Hu, J., Zhang, P., Chen, Q., Zhang, J., Li, T. & Wang, C.# Single-cell assignment using multiple-adversarial domain adaptation network with large-scale references. *Cell Reports Methods*, 3, 100577 (2023).

- Shi, X.*, Yu, Z.*, Ren, P., **Dong, X.**, Ding, X., Song, J., Zhang, J., Li, T.# & Wang, C.# HUSCH: an integrated single-cell transcriptome atlas for human tissue gene expression visualization and analyses. *Nucleic Acids Research*, 51, D1029-D1037 (2023).
- Xu, R.*, Li, S.*, Wu, Q.*, Li, C.*, Jiang, M.*, Guo, L., Chen, M., Yang, L., **Dong, X.**, Wang, H., Wang, C.#, Liu, X.#, Ou, X.# & Gao, S.# Stage-specific H3K9me3 occupancy ensures retrotransposon silencing in human pre-implantation embryos. *Cell Stem Cell*, 29, 1051-1066.e8 (2022).
- Sun, D.*, Wang, J.*, Han, Y.*, **Dong, X.**, Ge, J., Zheng, R., Shi, X., Wang, B., Li, Z., Ren, P., Sun, L., Yan, Y., Zhang, P., Zhang, F.#, Li, T.# & Wang, C.# TISCH: a comprehensive web resource enabling interactive single-cell transcriptome visualization of tumor microenvironment. *Nucleic Acids Research*, 49, D1420-D1430 (2021).
- Chen, C.-H., Zheng, R., Tokheim, C., **Dong, X.**, Fan, J., Wan, C., Tang, Q., Brown, M., Liu, J.S., Meyer, C.A. # & Liu, X.S.# Determinants of transcription factor regulatory range. *Nature Communications*, 11, 2472 (2020).
- Li, S.*, Wan, C.*, Zheng, R., Fan, J., **Dong, X.**, Meyer, C.A.# & Liu, X.S.# Cistrome-GO: a web server for functional enrichment analysis of transcription factor ChIP-seq peaks. *Nucleic Acids Research*, 47, W206–W211 (2019).

Manuscripts Under Revision and Review:

- **Dong, X.***, Ouyang Z.*, Liu H.*, Wang M.*, Yuan M., Sun D., Zhou X., Bai Z., Zhang Y., Lin C., Yang Y., Feng M., Wang H., Wang C.#, Gao Y#, Gao S# & Wei K.#, Spatiotemporal Multiomic Integration Reveals Critical Regulators of Cardiomyocyte Maturation and Proliferation (*Under Revision*)
- Zhu, X.*, Zhang, Y.*, Wang, N.*, **Dong, X.***, Tang, Z., Zheng, X., Ju, C.-W., Zhu, S., Fang, X., Liang, B., Qiu, Y., Ma, S., Jia, K., Ji, Y., Pan, Y., Xia, P., Feng, Q., Zhu, C., Li, H., He, C., Gao, Y.#, Si, J.#, Ge, Y.# & Liu, J.#, Base-resolution glycoRNA profiling using PRODAGS identifies abundant rRNA-derived fragments enhancing CAR-T efficacy (*Under Review*)
- Li, W.*, Ding, X.*, Zhao, Y.*, Si, W.*, **Dong, X.***, Zhang, L., Qiu, Y., Niu, J., Liu, L., Zheng, Z., Chen, Y., Ji, F., Xia, P., Ma, X., Mao, X., Zhang, B., Li, L., Zhao, M., Weichselbaum, R.R., Wang, L.# & He, C.#, Targeting YTHDF proteins with a small molecule elicits potent tumor-specific antitumor immunity (*Under Review*)

SKILLS

- **Bioinformatics**: Single-cell/spatial RNA/ATAC-seq analysis; Epigenome data analysis (ChIP-seq, ATAC-seq, etc.); Machine learning/Deep Learning (PyTorch, scikit-learn)
- **Programming**: Python; R; Shell; JavaScript; SQL
- **Computing:** Nextflow; Snakemake; AWS/GCP computation
- **Server operation and maintenance**: Ubuntu; CentOS / Rocky Linux

CONFERENCES

2026 Poster Presentation, AI and Biology, Cold Spring Harbor Conferences Asia, Suzhou, China

2024 Poster Presentation, Human Cell Atlas Asia 2024, Hong Kong SAR, China

2024 Poster Presentation (Awarded), The 2nd Bioinformatics Conference of Yangtze River Delta, Shanghai, China

2023 Poster Presentation, Systems biology of gene regulation and genome editing, Cold Spring Harbor Conferences Asia, Suzhou, China

2023 Poster Presentation, The WLA Research Conference on Cells and Genes, Shanghai, China

2023 Poster Presentation, The 11th National Conference on Bioinformatics and Systems Biology, Guangzhou, China

2021 Poster Presentation, Human Cell Atlas Asia 2021 Meeting, Online

2021 Poster Presentation, The 10th National Conference on Bioinformatics and Systems Biology, Chengdu, China

TEACHING

Teaching Assistant, Summer Deep Learning Course, School of Life Sciences and Technology, Tongji University (2023, Shanghai)

Teaching Assistant, Summer Bioinformatics Course, School of Life Sciences and Technology, Tongji University (2020-2022, Shanghai)

Teaching Assistant, Dragon Star Course, The 14th International Bioinformatics Workshop (2019, Beijing)

AWARDS AND DISTINCTIONS

2024 Outstanding Graduate Awards of Shanghai, Shanghai Municipal Education Commission

2023 The 17th Academic Pioneer (Top 0.05%), Tongji University

2023 National Scholarship (Top 0.2%), Ministry of Education of the People's Republic of China

2022 Outstanding Student Leader, Tongji University

2019 Outstanding Graduate Student Scholarship, Tongji University

2018 Outstanding Undergraduate Thesis for Bachelor's Degree, Shandong Normal University

2017 Outstanding Graduate Awards of Shandong, Human Resources and Social Security Department of Shandong Province

2017 The First Prize of the 15th Challenge Cup Competition in Shandong, Shandong Association for Science & Technology

2017 Outstanding Student Leader of Shandong, Shandong Provincial Education Department

2016 The Grand Prize of Shandong College Students' Biochemistry Experiment Skills Competition, Shandong Association of Science & Technology